

Adroit Technologies

Protocol Driver

Name	Strike ENERMAX Plus
DLL	STKEMAXP



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Contents

1. Overview	4
2. Wiring	5
3. Device Configuration	6
3.1. Device Configuration	6
3.1.1. TCP/IP Settings	7
3.1.2. Modem Settings.....	7
4. Supported Addresses	9
5. Protocol Definition and Considerations	12
6. General Notes and Troubleshooting	13
6.1. Contact Details.....	13
6.2. Adroit Technologies Knowledge Base.....	13
7. Appendix A - Profile Modifiers	14
8. Revisions	15
8.1. Document Revision.....	15
8.2. DLL Revision	15

1. Overview

Adroit supports communication with the following Strike Devices:

- Enermax+ (RS-232C or Ethernet TCP/IP)

2. Wiring

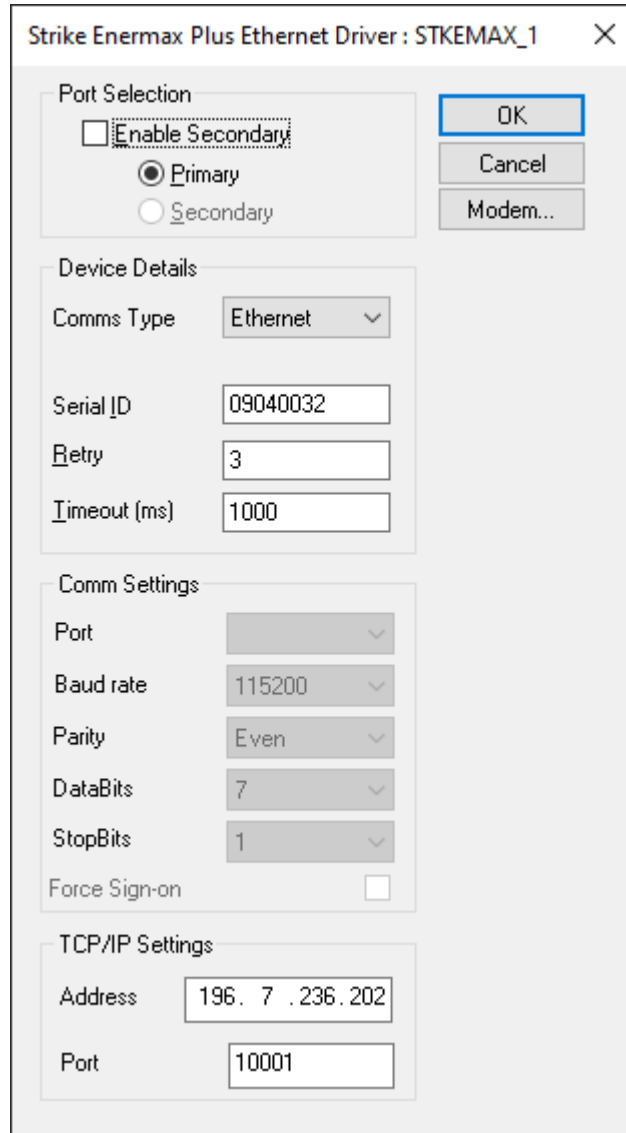
Standard RS-232C cable (DTE-DTE) cross-over cable with or without handshaking.
The driver expects the communication link to be point-to-point due to the session-based protocol used.

OR

Standard Ethernet UTP Cable.

3. Device Configuration

3.1. Device Configuration



The image shows a Windows-style dialog box titled "Strike Enermax Plus Ethernet Driver : STKEMAX_1". It contains several sections for configuring the device:

- Port Selection:** Includes a checkbox for "Enable Secondary" (unchecked), a radio button for "Primary" (selected), and a radio button for "Secondary" (unchecked).
- Device Details:** Includes a "Comms Type" dropdown menu set to "Ethernet", a "Serial ID" text box containing "09040032", a "Retry" text box containing "3", and a "Timeout (ms)" text box containing "1000".
- Comm Settings:** Includes dropdown menus for "Port", "Baud rate" (set to "115200"), "Parity" (set to "Even"), "DataBits" (set to "7"), and "StopBits" (set to "1"). There is also a "Force Sign-on" checkbox (unchecked).
- TCP/IP Settings:** Includes an "Address" text box containing "196. 7 .236.202" and a "Port" text box containing "10001".

Buttons on the right include "OK" (highlighted with a blue border), "Cancel", and "Modem...".

Comms Type – Select communications type:

Select 'Serial' for Serial communications.

Select 'Ethernet' for Ethernet communications.

Serial ID – Serial identification number unique to the meter.

Retry - The number of times that Adroit will try to gain communication with the meter before failing a transaction.

Timeout – The time that Adroit will wait for a valid response from the meter before failing a transaction (Read or Write). Ensure proper timeout if using modem connection as it is much slower, where the maximum is 9600bps.

Modem – Opens the Modem Dialog if using Modem Connection.

Port – The available serial ports on your machine.

Baud rate – (Default = 115200) Select a baud rate: 9600, 19200, 38400, 56000, 57600, 115200

Parity – (Default = Even)

DataBits – (Default = 7)

StopBits – (Default = 1)

Force Sign-on – (Default = unchecked) If checked the driver will force a sign on and off for every transaction used when having multiple serial devices on a bus network.

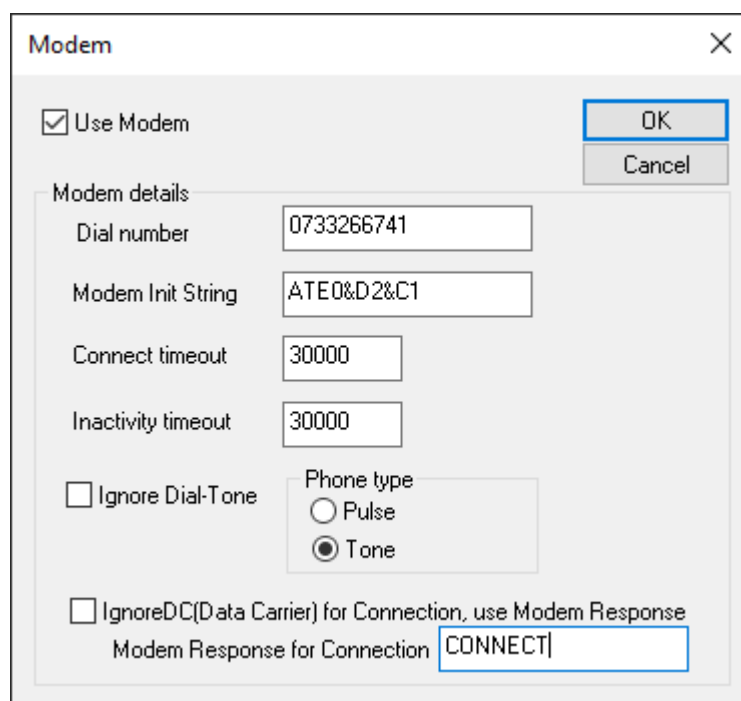
WARNING: communications will be slower.

3.1.1. TCP/IP Settings

Address – IP address of the meter.

Port – (Default = 10001) The port configured for the meter.

3.1.2. Modem Settings



The image shows a 'Modem' settings dialog box. It has a title bar with a close button (X). Inside, there is a checkbox labeled 'Use Modem' which is checked. To the right of this checkbox are 'OK' and 'Cancel' buttons. Below the checkbox is a section titled 'Modem details'. This section contains several fields: 'Dial number' with the value '0733266741', 'Modem Init String' with the value 'ATE0&D2&C1', 'Connect timeout' with the value '30000', and 'Inactivity timeout' with the value '30000'. There is also an unchecked checkbox labeled 'Ignore Dial-Tone'. To the right of this checkbox is a 'Phone type' section with two radio buttons: 'Pulse' (unchecked) and 'Tone' (checked). At the bottom, there is an unchecked checkbox labeled 'IgnoreDC(Data Carrier) for Connection, use Modem Response'. To the right of this checkbox is a text field labeled 'Modem Response for Connection' containing the text 'CONNECT|'. The 'OK' button is highlighted with a blue border.

Use Modem – Enable/Disable the Use of Modem on RS232.

Dial Number – Number to dial that is situated at end device.

Modem Init String – String to Initialize Local Modem with before dialling.

Connect Timeout – Time(ms) to wait for connection.

Inactivity Timeout – Time(ms) to wait where no traffic is flowing over modem connection before terminating the connection.

Ignore Dial Tone – Enable/Disable the existence of a dial-tone on the modem.

Tone/Pulse – If Dial Tone exists then it can be Tone or Pulse dial tone.

Ignore DC for Connection, use modem response – When DC(Data Carrier) is not wired or Virtual Serial Port device does not support it, then enable this option where we can only know a successful connection by modem response.

Modem Response for connection – Response modem will give when connection is established.

4. Supported Addresses

Enermax plus data address types and maximum ranges supported by the driver.

Area	Min Range	Max Range	String	Integer	Real	Read	Write
I Area (instantaneous)*1	0	39	-	-	I:1	Y	N
D Area (Display)*2	17	65535	D1:86	D2:86	D2:86	Y	N
P Area (Profile)*3	0	7	-	-	P:0 P:0 M100 P:0 M0.01 S2 P:0 S400	Y	N
A Area (Admin)*4	0	4	-	A:0	-	Y	Y

***1** Direct addresses for instantaneous read (I Area) values I:x where x = 0 .. 39...

- 0 - float - Volts_L1
- 1 - float - Volts_L2
- 2 - float - Volts_L3
- 3 - float - Amps_L1
- 4 - float - Amps_L2
- 5 - float - Amps_L3
- 6 - float - Watts_L1 (Multiplied by (1 / Accumulator))
- 7 - float - Watts_L2 (Multiplied by (1 / Accumulator))
- 8 - float - Watts_L3 (Multiplied by (1 / Accumulator))
- 9 - float - Watts_total (Multiplied by (1 / Accumulator))
- 10 - float - Vars_L1 (Multiplied by (1 / Accumulator))
- 11 - float - Vars_L2 (Multiplied by (1 / Accumulator))
- 12 - float - Vars_L3 (Multiplied by (1 / Accumulator))
- 13 - float - Vars_Total (Multiplied by (1 / Accumulator))
- 14 - float - VA_Harmonic_L1
- 15 - float - VA_Harmonic_L2
- 16 - float - VA_Harmonic_L3
- 17 - float - VA_Total_Harmonic
- 18 - float - VA_L1 (Multiplied by (1 / Accumulator))
- 19 - float - VA_L2 (Multiplied by (1 / Accumulator))
- 20 - float - VA_L3 (Multiplied by (1 / Accumulator))
- 21 - float - VA_Total (Multiplied by (1 / Accumulator))
- 22 - float - Accumulator
- 23 - float - Quadrant_L1
- 24 - float - Quadrant_L2
- 25 - float - Quadrant_L3
- 26 - float - Quadrant_Total
- 27 - float - PF_RV_I (Multiplied by 0.001)
- 28 - float - PF_YV_I (Multiplied by 0.001)
- 29 - float - PF_BV_I (Multiplied by 0.001)
- 30 - float - PF_TV_I (Multiplied by 0.001)
- 31 - float - PHI_RV_I (Multiplied by 0.1)
- 32 - float - PHI_YV_I (Multiplied by 0.1)
- 33 - float - PHI_BV_I (Multiplied by 0.1)
- 34 - float - PHI_TV_I (Multiplied by 0.1)
- 35 - float - PHI_RV_YV (Multiplied by 0.1)
- 36 - float - PHI_RV_BV (Multiplied by 0.1)
- 37 - float - Frequency (Multiplied by 0.01)
- 38 - float - Internal_Temp_c (Multiplied by 0.1)
- 39 - float - Internal_Temp_f (Multiplied by 0.1)

- *2) The display reads numbers can be found in Emaxplus-CM-RDU-3rd-Party-0v01.xls
Please remember that Dx:y , x is the row and you usually want to scan D2:y not 1 , which will give you the display name... but it can be scanned if you really want to... just use D1:y
- *3) 8 profile channels available as floating-point values, only Real/Analog agents will be supported for profile data.



WARNING

Channels are dynamically ordered from the smallest period to the largest. If you add a channel or edit a channel in the meter, the SCADA MUST take this into account that channels positions could move.

Example: channel 1 period = 10 minute sample rate, channel 2 = 30 minute sample rate, now you add channel 3 = 5 minute rate, then channel 3 will become channel 1 in memory, and the old 1 and 2 will become 2 and 3 respectively. So, if channel 1 was VOLTS, and channel 3 was VARS, the next profile read, channel 1 would have VARS instead of VOLTS. This is like swapping two digital jumpers on a digital output card without telling the SCADA engineer to update the user interface – a wrong indication is presented.



WARNING

Channels cannot be skipped in the read. So, for example, if channel 2 were to be deleted from the meter, the remaining channels would move down the list in this case channel 3 would move to channel 2.

Profile modifiers: Modifiers can be added to a channel to modify the value before it is sent to Adroit.

You can multiply(divide) a value and/or subtract(add) a value. (See Profile Modifiers Appendix.)

- *4) These are administrative registers used in advanced function that may require more than one type of data to be present before execution can take place, for example profile retrieval of multiple days.

Admin area consists of 5 16bit registers only available to integer type agents such as Integer/Marshal.

Register 0 (Address A:0) is not persisted data, will be reset when the Agent Server restarts.

Register 1-4 (Address A:1-4) are persisted, so these will be preserved between Agent Server restarts, and there is no need to save WGP file manually, the data is persisted to the device config in the registry.

Register 0 [bit packed 16 (unpersisted)]

- { Bit 0 = Do time synchronisation (>20min will cause a fast set)
- Bit 1 = Do profile read (will use Registers 2 and 3 for required parameters)
- Bit 2 ... Bit 15 = not used yet.}

Register 1 [bit packed 16 (persisted)]

- { Bit 0 = Enable "fetch yesterday" profile data (uses Register 4 as time in minutes from 00:00 to start)
- Bit 1... Bit 15 = not used yet.}

Register 2 [unsigned short] = day offset from today for profile read (see bit 1 of register 0) (1 = yesterday) (0 = today) (>1 days further into the past)

Register 3 [unsigned short] = count of number of days to fetch from offset (Register 2)

Examples:

Reg2 = 1, Reg3=2, Reg0.1=1 (**NB** write Reg0 last) will result in a profile read for yesterday and today up to the last recorded value.

Reg2 = 30, Reg3=15, Reg0.1=1 will result in a profile read from 30days ago for 15 days.

Register 4 [unsigned short] = time in minutes to add to 00:00 as when to start a fetch yesterday.

Examples:

Reg4 = 120, Reg1.0 = 1 will result in a profile read to "fetch yesterday's data" at 2:00am every morning.

Remember Reg1.0 will be persisted.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Appendix A - Profile Modifiers

Each channel's address can be postfixed with a modifier to allow for manipulation of the value before it is sent and stored in Adroit. This is used to convert the profile reading from Raw values to True Value readings, additional ratios (CT/VT) could be added.

Only 1 multiplier is allowable so True value factor must be merged with other multiplication requirements such as VT ratio etc.

Additionally, only 1 subtractor is allowable.

(Subtraction takes place after multiplication, if a multiplier is specified, else if only the subtractor is specified only that will be applied.)

TVF – True Value Factor, CT – Current Ratio, VT – Voltage Ratio.

Example: Profile 1 modified by TVF 0.01 (division by 100)

"P:1 M0.01"

P:n Mv – (P)rofile : (n)umber (M)ultiply by (v)alue.

Example Profile 2 modified by TVF 0.01 and VT 10

$0.01 \times 10 = 0.1$ total.

"P:1 M0.1"

P:n Mv – (P)rofile : (n)umber (M)ultiply by (v)alue.

Example Profile 3 modified by TVF 0.01 and subtract 400.

"P:1 M0.01 S400"

P:n Mv Sv– (P)rofile : (n)umber (M)ultiply by (v)alue, finally (S)ubtracting a (v)alue.

See below for current modifiers as documented by Strike.

Slot Type	Derived Category	TVF	Maximum Float: Scaled Integer
Phase angle	x10	0.1	360.0 : 3600
Power factor	x1000	0.001	2.000 : 2000 (sign offset +1)
Quadrant	x1	1	4 : 4
Frequency	x10	0.1	65.0 : 650
Temperature	x10	0.1	800.0 : 8000 (sign offset +400)
Voltage	x10	0.1	660.0 : 6600
Current	10A : x1000 160A : x100	10A : 0.001 160A : 0.01	10.000 : 10000 160.00 : 16000
Power/Energy	10A : x1 160A : /10	10A : 1 160A : 10	8250.0 : 8250 132000 : 13200

(Aside from applying the TVF)

Slot Type	To Do	Display Format
Phase angle	Nothing extra	xxx.x
Power factor	Subtract 1.0	±x.xxx
Quadrant	Nothing extra	X
Frequency	Nothing extra	xx.x
Temperature	Subtract 400.0	±xxx
Voltage	Allow user to choose if VT ratio to be applied	xxxxxx.x
Current	Allow user to choose if CT ratio to be applied	xxxx.xxx
Power/Energy	Allow user to choose if IMC to be applied	Use demand category scaling

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	13-Jan-2023	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Grammar edits, formatting changes and content additions	D. Jordaan
2.0				
3.0				

8.2. DLL Revision

Rev.	Date	Changes
2	01/07/2009	Enabled serial support.
3a	10/09/2009	added documentation regarding P and A registers.
3d	06/10/2009	added documentation regarding P modifiers.
5	02/07/2010	added modem support.
12	18/01/2012	Extra check on protocol to detect out of sequence messaging.

